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隱形帳簿：量化印尼平台經濟中的隱形楔子
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

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Abstract

Digital economy is measured through platform revenue, national e-commerce statistics and payment data, which are often used interchangeably even though each records a different part of the same activity. This study asks how much economic activity remains invisible when digital platforms are measured through their reported revenue. I define the invisible wedge as the difference between transaction value and platform-recognized revenue, measure it across Indonesian platform histories from 2020 to 2025, and compare the result with BPS-Statistics Indonesia e-commerce data and Bank Indonesia payment records. In 2023, three major platforms processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue, a ratio of 13.7 to 1. Across twelve year-to-year comparisons, revenue grew faster than transaction value in ten, so the relationship is not stable. National statistics place 98.5 percent of recent e-commerce growth outside the marketplace component, and payment measures grew up to eleven times as fast as e-commerce value. Eight platform businesses outside Indonesia provide external corroboration. The proposal treats these records as distinct measurement layers and tests whether substituting one for another changes conclusions about the scale and growth of digital activity.

1. Introduction

Southeast Asia's digital economy grew from approximately US\$100 billion in 2020 to US\$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). Digital platform scale, however, is not the same thing as platform revenue. Platforms record transactions, determine participant payouts, and book their own revenue, while national statistics and payment systems record other parts of the same economic activity. These measures are often used interchangeably even though they do not measure the same thing. This paper argues that doing so produces systematically wrong conclusions about the scale and growth of platform economies.

Suppose a platform processes 100 units of transaction value but recognizes only 10 as revenue. The remaining 90 are still recorded within the platform's system but do not constitute platform revenue; they represent merchant receipts, driver payouts, inventory costs, taxes, and other pass-through payments. An analyst using the platform's accounts sees 10, while one measuring the commerce it processes sees 100. I call the difference the invisible wedge.

Indonesia, the largest digital market in Southeast Asia, provides a clear empirical setting because several records of the same activity can be compared. Platform companies disclose transaction and revenue information; BPS-Statistics Indonesia, the national statistical agency, reports e-commerce activity; and Bank Indonesia, the country's central bank, reports digital payments. In 2023, the platform comparison covers Shopee and Tokopedia, which together represented about 70 percent of estimated Indonesian marketplace transaction value, together with Grab, a major intermediary in delivery and other digital services. The three cases processed US\$43.23 billion of transaction value against US\$3.16 billion of recognized revenue, leaving a US\$40.07 billion invisible wedge.

The difference also changes over time. Across twelve year-to-year comparisons between 2020 and 2025, revenue grew faster than transaction value in ten, with a median absolute gap of approximately 42 percentage points between the two growth rates; in some years the two measures moved in opposite directions. Platform revenue therefore understates the level of platform-mediated commerce while, in most observed years, overstating its growth.

The same problem appears beyond company accounts. BPS reports that national e-commerce value grew 17.1 percent between 2023 and 2024, but about 98.5 percent of that increase occurred outside the marketplace component; digital payment measures reported by Bank Indonesia grew substantially faster still. Indonesia's Minister of Finance Regulation No. 37 of 2025, which introduces seller reporting and withholding through designated marketplaces, provides a further test of how platform-held transaction records enter administrative reporting.

1.1 Research question and objectives

One question drives the proposal:

How much economic activity remains invisible when digital platforms are measured through their reported revenue?

Indonesia provides the main empirical setting. Three objectives follow: first, to measure the invisible wedge across available platform histories from 2020 to 2025; second, to explain large movements using disclosed changes in monetization, customer incentives, revenue recognition, and reporting scope; and third, to compare the platform evidence with national e-commerce and payment records to determine whether they produce the same account of economic scale and growth.

A separate comparison of eight platform businesses outside Indonesia tests whether the same transaction-revenue separation appears under other business models.

1.2 Contributions

First, the study develops the invisible wedge from a one-year estimate into a measure that can be followed over time. From 2020 to 2025, revenue grows faster than transaction value in ten of twelve annual comparisons, showing that platform revenue is not a stable measure of either the level or the growth of platform-mediated commerce.

Second, I examine why the wedge moves using accounting components disclosed by the platforms themselves. Between 2022 and 2023, Tokopedia's transaction value fell 8.9 percent while its third-party net revenue rose 53.2 percent. About 60.6 percent of the arithmetic increase in net revenue is associated with lower customer incentives, with the remainder associated with higher gross revenue.

Third, I extend the comparison beyond company accounts. National e-commerce statistics show that almost all of Indonesia's 2023–2024 growth occurred outside the marketplace component, while digital payment measures grew substantially faster than e-commerce value. The records therefore produce different conclusions about the scale, growth, and location of digital activity.

Fourth, I test whether the main findings survive alternative assumptions, sample rules, and external platform evidence rather than depending on one preferred construction.

1.3 Positioning the contribution

The contribution is a finance measurement problem before it is a tax-policy one. Platform accounts make the intermediation boundary observable, but that boundary affects the conclusions investors, researchers, statistical agencies, and policymakers draw about scale, growth, and monetization. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter, while Berg et al. (2020) show that intermediary-held digital traces contain information conventional records can miss. This paper applies those ideas to platform economies and shows that the measure chosen to describe digital activity can change the economic conclusion.

2. Literature Review and Research Gap

Digital platforms sit between economic activity and the records used to measure it. They facilitate transactions between buyers and sellers, recognize only part of that activity as their own revenue, and generate records that can also appear in payment, statistical, and administrative systems. Existing research explains each part of this process from a different angle. Taken together, it provides the foundation for asking how much activity remains invisible when platforms are measured through reported revenue.

2.1 Platform Economics and Revenue Recognition

Platform economics explains why the value of commerce processed through a platform can be much larger than the revenue the platform records. Digital platforms often connect consumers with merchants or other service providers and earn commissions, fees, advertising, logistics, or other intermediation revenue rather than the full value of every transaction they facilitate (Rochet and Tirole 2003; Caillaud and Jullien 2003; Parker and Van Alstyne 2005; Armstrong 2006; Hagiu and Wright 2015; Evans and Schmalensee 2016).

Accounting can widen or narrow this difference. Under IFRS 15, whether a company acts as a principal or an agent affects whether it recognizes the full customer payment or only the amount it earns from arranging the transaction (International Accounting Standards Board 2014). Customer incentives, monetization, service mix, acquisitions, and changes in business scope can also move reported revenue without an equivalent change in transaction activity. De Franco, Kothari and Verdi (2011) show why comparable accounting bases matter when financial information is interpreted. The literature therefore explains why a transaction–revenue gap can exist and move over time, but not how economically large that gap becomes in practice.

2.2 Digital Records and Third-Party Information

Activity outside platform revenue is not necessarily unrecorded. Merchants and service providers can be small, self-employed, or weakly represented in conventional financial records while their transactions still leave detailed records inside a digital platform. This differs from the usual shadow-economy problem, where part of the difficulty is that economic activity leaves few reliable formal records (La Porta and Shleifer 2014; Ulyssea 2018; Medina and Schneider 2019). Digital records can contain useful economic information that conventional records miss (Berg et al. 2020), and platform-mediated activity can create additional financial and administrative traces (Barrios, Hochberg and Yi 2022; Denes, Lagaras and Tsoutsoura 2025).

These records also matter once they become available to government. Public-finance research shows that compliance and enforcement change when information about income or transactions is independently reported by third parties (Kleven et al. 2011; Pomeranz 2015; Naritomi 2019; Kleven, Kreiner and Saez 2016; Slemrod 2019). The OECD Model Rules (OECD 2020) and the European Union's DAC7 regime (European Union 2021) apply this principle to digital platforms. Indonesia's Minister of Finance Regulation No. 37 of 2025 follows the same logic through seller identification, transaction-linked reporting, and marketplace withholding. The relevance here is informational: transaction records outside platform revenue may still exist and become usable elsewhere.

2.3 Measuring the Digital Economy

Official measurement frameworks already distinguish between the commerce facilitated through a platform and the service provided by the platform itself. National-accounting guidance separates the buyer-seller transaction from the platform's intermediation service rather than treating the full transaction value as platform output (Ahmad and Schreyer 2016; International Monetary Fund 2018; OECD 2023; United Nations et al. 2025).

In practice, however, gross transaction value, platform revenue, e-commerce statistics, and payment data are all used to describe digital economic activity. They measure different things. A difference that is harmless when the measures move together becomes important when they do not, because the choice of measure can change conclusions about economic scale and growth. This thesis therefore compares the records rather than assuming that one can substitute for another.

2.4 Research Gap

Existing research explains the pieces of the problem separately. Platform economics explains why transaction value can exceed revenue; accounting explains why the gap can move; research on digital records explains why activity outside revenue can still be observable; and official measurement frameworks explain why these quantities should not be treated as equivalent.

What remains less clear is how large these differences become in practice, whether they persist over time, and whether choosing one measure instead of another changes the economic conclusion. This thesis addresses that gap by measuring the invisible wedge over time, examining disclosed reasons for major movements, comparing platform accounts with national e-commerce and payment records, and using external platform evidence to test whether the same measurement problem appears beyond the main Indonesian setting.

3. Theoretical Framework and Hypotheses

Digital platform activity can be observed through several records, but those records do not measure the same economic object. The framework begins with the difference between the value a platform processes and the revenue it recognizes, then asks whether that relationship remains stable over time and whether other records produce the same account of digital activity.

For the platform analysis, transaction value and revenue are paired only when they refer to the same period and business activity. Broader-scope, reconstructed, national-statistical, and payment evidence is kept in separate analytical roles rather than combined into one sample.

3.1 Measurement Framework

Let V_{it} denote transaction value processed by platform i in period t , and R_{it} the revenue recognized over the same matched period and business scope. The absolute invisible wedge is:

$$W_{it} = V_{it} - R_{it}$$

W measures transaction value outside the platform's recognized revenue. It does not represent merchant profit, worker income, taxable income, unpaid tax, tax evasion, or value added.

Because platforms differ in size, I also express the wedge relative to revenue. The Ecosystem Ratio is:

$$E_{it} = (V_{it} - R_{it}) / R_{it}$$

An E of 9 means that for every one unit of platform revenue, nine additional units of transaction value lie outside the revenue line. The Ecosystem Ratio is constructed for this study rather than adopted from prior work. Its basis is the platform-economics result that booked revenue need not track the transaction value a multisided platform coordinates (Rochet and Tirole 2003; Hagiu and Wright 2015; Evans and Schmalensee 2016), together with the accounting result that the share of transaction value entering revenue is itself a reporting choice (International Accounting Standards Board 2014; De Franco, Kothari and Verdi 2011).

A single ratio does not show whether transaction value and revenue move together through time. I therefore define annual growth divergence as:

$$D_{it} = g(V_{it}) - g(R_{it})$$

A positive D means transaction value grew faster than revenue; a negative D means revenue grew faster. Together, W , E , and D measure the size of the transaction-revenue boundary and how that boundary changes over time.

3.2 Hypotheses

The four hypotheses examine different layers of the same measurement problem. H1 tests the transaction-revenue boundary within platforms. H2 and H3 examine where activity and records outside platform revenue appear in national e-commerce evidence. H4 asks whether the different records ultimately produce different conclusions about digital economic activity. The evidence layers are analyzed separately, and none of the hypotheses is interpreted as a causal claim. Table 1 pairs each hypothesis with the evidence that tests it.

H1. Transaction value and platform revenue diverge materially within the same platform over time.

If revenue were a stable proxy for the commerce processed by a platform, transaction value and revenue would move proportionally.

H2. Aggregate e-commerce growth partly reflects changes in the number of participating businesses rather than only changes in transaction value per business.

This separates changes associated with wider business participation from changes in the average value observed per business.

H3. Marketplace participation is associated with stronger financial recordkeeping.

This asks whether marketplace-mediated activity is associated with a different recordkeeping pattern from other e-commerce activity. Association does not imply that marketplace participation causes formalization.

H4. Measurement choices materially alter conclusions about the scale and growth of digital economic activity.

This asks whether platform accounts, national e-commerce statistics, and payment records produce the same economic account when they observe different parts of digital activity.

Table 1. Hypotheses and principal empirical tests.

Hypothesis	Principal empirical test
H1 - Transaction-revenue divergence	Matched platform transaction value and revenue over time, the Ecosystem Ratio, and annual growth divergence
H2 - Sources of e-commerce growth	BPS national e-commerce value and business-count decomposition
H3 - Marketplace participation and recordkeeping	BPS published marketplace and financial-recordkeeping evidence
H4 - Measurement choice	Comparison of platform, national e-commerce, and payment records

The separate comparison of platform businesses outside Indonesia is used as corroboration rather than as a fifth hypothesis. Indonesia's marketplace reporting regime likewise remains an institutional application. Whether platform-held records become administratively linked to seller obligations requires post-implementation evidence and cannot be established from the legal framework alone.

4. Data and Methodology

The study uses several sources because no single record captures every part of digital platform activity. Platform reports provide transaction value and revenue; BPS-Statistics Indonesia provides national e-commerce and business evidence; Bank Indonesia provides payment statistics; and platform companies outside Indonesia provide a separate comparison. These sources answer different questions and are not combined into one dataset.

4.1 Selecting the Platform Evidence

The basic observation is one platform in one year, with transaction value and revenue referring to the same period and, as closely as possible, the same business activity.

An observation is retained only when its period, geographic and business coverage, units, definitions, and source can be identified. Measures referring to different periods are excluded, while major changes in accounting definitions or business structure are recorded.

Table 2. Indonesian platform evidence.

Type of evidence	Platforms	Annual observations	Role
Direct Indonesia-aligned data	Tokopedia e-commerce	2	Main direct anchor
Company-reported data with broader scope	Blibli third-party business; Bukalapak Group	9	Evidence through time with stated scope limits
Indonesia estimates requiring construction	Grab; Shopee Indonesia	6	Supporting country evidence

Table 2 groups the retained evidence by how directly it maps to Indonesia. An annual observation means one matched transaction-and-revenue pair for one platform in one year, and the 17 retained observations produce 12 year-to-year comparisons because the same platforms are not available in every year.

Tokopedia is closely aligned with Indonesia but is not a literal country line. Blibli includes online travel, Bukalapak includes some overseas activity, and Grab and Shopee require additional construction.

4.2 Comparing Platform Activity Through Time

Changes through time are examined mainly by comparing each platform with itself. This keeps the company, business activity, and reporting convention as similar as the disclosures allow.

When transaction value and revenue move sharply apart, disclosed changes in monetization, customer incentives, revenue recognition, and business scope are used to reconcile the movement where the filings permit. These reconciliations are arithmetic accounting explanations rather than estimates of causal effects.

Companies use different names for transaction activity, including gross transaction value (GTV), gross merchandise value (GMV), and total payment volume (TPV). This proposal uses GTV as its common label for the value of transactions processed through a platform, and each issuer's own label and definition are retained and checked rather than assumed to be identical.¹

A separate comparison uses the 2023 fiscal year (FY2023) because it provides the most useful common year across the principal platform cases. It shows the scale of the transaction-revenue difference among the selected cases, not the size of Indonesia's entire digital economy. Market-share estimates describe transaction-value coverage only, not wider representativeness.

4.3 Direct and Constructed Indonesia Measures

Not every company reports the same Indonesia-specific information. Table 3 sets out, for each platform, what is disclosed directly, what has to be constructed, and the limitation that follows.

Table 3. Construction of the principal platform measures.

Platform	Directly available	Constructed or external input	Main limitation
Tokopedia	E-commerce transaction value and segment revenue	None for matched pair	Indonesia-aligned segment, not literal country line
Grab	Indonesia revenue; Group revenue and GTV	Indonesia GTV derived from Group figures	Assumes Group revenue-to-GTV relationship applies to Indonesia
Shopee	Group service monetization rate	External Indonesia transaction estimate; Indonesia revenue derived	Country measures are not independent disclosures
Blibli / Bukalapak	Reported transaction and revenue pairs	None for retained pairs	Scope extends beyond Indonesia alone

Grab reports Indonesian revenue directly but not Indonesian transaction value. Indonesian GTV is therefore estimated as:

$$\text{Indonesia GTV} = \text{Indonesia revenue} \times \text{Group GTV} / \text{Group revenue}$$

The Group monetization rate is platform revenue divided by gross transaction value, and the result is treated as a reconstruction rather than a direct disclosure.

Sea Limited does not report Shopee transaction value separately for Indonesia. Indonesian transaction value is taken from Momentum Works, a Singapore-based industry research firm whose annual Southeast Asian e-commerce report is widely cited for country-level platform transaction estimates. It is used because no issuer discloses Shopee's Indonesian transaction value, it is labeled as a third-party estimate wherever it appears, and Indonesian revenue is then estimated using Sea's Group service monetization rate:

$$\text{Indonesia revenue} = \text{Indonesia transaction value} \times \text{Group monetization rate}$$

Both constructions remain explicitly labeled wherever they are used.

¹ Issuer labels differ. GoTo reports GTV and Sea Limited reports GMV for Shopee; for these the label differs but the measure is the same. Blibli and Bukalapak report TPV for the segments used here, and Grab reports GMV for deliveries and mobility and TPV for financial services. Where a source reports TPV the measure rests on a different basis, so it is retained under its own label rather than relabeled as GTV.

4.4 Evidence Beyond Platform Accounts

Table 4. Evidence modules and their role.

Evidence source	What is observed	Role
Indonesian platform histories	Transaction value and revenue through time	Measure the wedge and its movement
BPS national e-commerce statistics	National transaction value and estimated number of businesses	Examine sources of aggregate growth
BPS marketplace study	Business-level marketplace and recordkeeping evidence	Examine the association with financial recordkeeping
Bank Indonesia	Digital-payment measures	Compare payment growth with e-commerce growth
Platform companies outside Indonesia	Matched transaction and revenue measures	Test whether the same boundary appears elsewhere

Table 4 gives each source a single role. The modules answer different questions, so they are reported separately rather than pooled into one estimate.

For H2, national e-commerce value is expressed as:

$$V = N \times A$$

where N is the estimated number of e-commerce businesses and A is implied nominal transaction value per business. Changes in total value are divided arithmetically between the business-count and value-per-business components. This does not identify causal entry, productivity, or growth among continuing firms. A conflicting published 2023 business count is retained as a sensitivity check.

For H3, BPS's published business-level evidence is used to examine the association between marketplace participation and financial recordkeeping. Province-level correlations are supporting diagnostics only.

Bank Indonesia payment measures provide a separate comparison for H4. They are compared with e-commerce growth but are not treated as sales estimates or added to platform or BPS transaction values.

4.5 External Evidence and Robustness

The external comparison contains 48 matched annual transaction-and-revenue observations from eight platform businesses covering 2017-2025. It tests whether the same transaction-revenue boundary appears under other business models; it is not a representative sample of global e-commerce and does not enlarge the Indonesian sample.

Robustness checks vary admission rules, revenue definitions, publication versions, and assumptions used in reconstructed country measures. The resulting ranges are treated as sensitivity or scenario ranges rather than statistical confidence intervals.

5. Preliminary Evidence and Feasibility

The evidence assembled so far is sufficient to establish that the proposed comparisons can be carried out and that the choice of measurement produces economically meaningful differences. The preliminary results first show the scale and movement of the transaction-revenue boundary, then examine whether the wider Indonesian evidence tells the same story.

5.1 Platform Scale and Movement

The 2023 fiscal year provides a common comparison across Grab, Tokopedia, and Shopee.

Table 5. Indonesia-focused 2023 platform comparison, US\$ billions.

Case	Transaction value V	Revenue R	Wedge W	Ecosystem Ratio E	Evidence
Grab Indonesia	5.381	0.605	4.776	7.895x	Transaction value reconstructed
Tokopedia	16.331	0.405	15.926	39.296x	Direct matched pair
Shopee Indonesia	21.520	2.152	19.368	9.000x	Transaction value and revenue estimated
Selected cases	43.233	3.162	40.070	12.671x	Sum of three cases, not national total

Table 5 reports that cross-section. Across the three cases, US\$43.23 billion of transaction value corresponds to US\$3.16 billion of recognized platform revenue, leaving a US\$40.07 billion wedge, so transaction value is approximately 13.7 times recognized revenue. For scale, that wedge is equivalent in magnitude to about 2.9 percent of Indonesia's 2023 nominal GDP of US\$1.371 trillion. The comparison is one of magnitude only: the three platforms are not the whole market, and the wedge is transaction value outside platform revenue, not value added.

Shopee's ratio of 9.0 follows from the Group monetization rate used to estimate its Indonesian revenue. In preliminary checks, rates between 9 and 11 percent move the combined wedge only between US\$39.86 billion and US\$40.29 billion, and excluding any one platform leaves a wedge of at least US\$20.70 billion.

The longitudinal evidence shows that this relationship is not constant. Across twelve year-to-year comparisons, revenue grows faster than transaction value in ten, with a median absolute growth divergence of 42.02 percentage points.

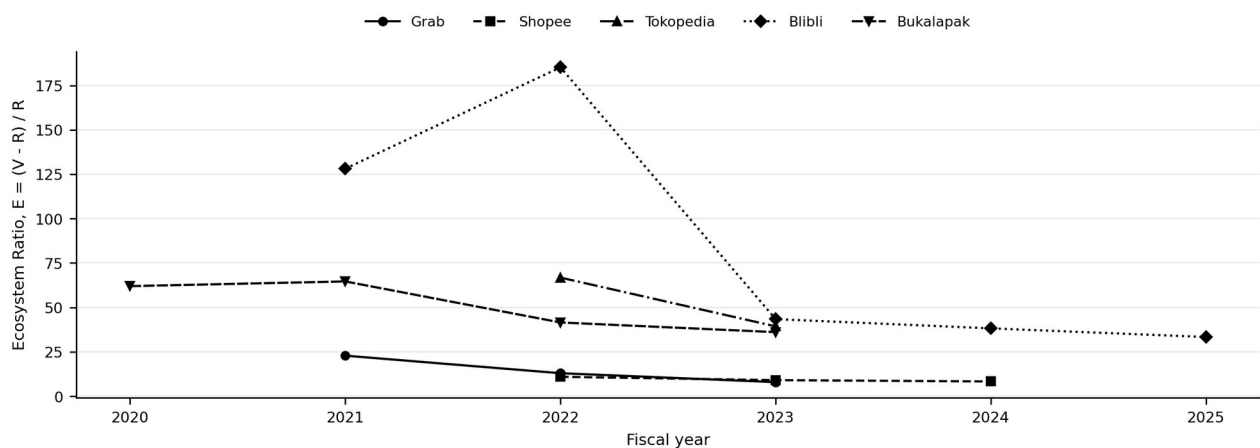


Figure 1. Ecosystem Ratio through time for each retained platform. Each line follows one platform's matched figures; differences between platforms are descriptive.

Tokopedia provides the clearest illustration of why the two measures can separate. Between 2022 and 2023, transaction value falls 8.9 percent while third-party net revenue rises 53.2 percent. The change in net revenue can be reconciled arithmetically to lower customer incentives and higher gross revenue rather than treated as an unexplained residual.

5.2 Evidence Beyond Platform Revenue

National e-commerce and business evidence provides a different view of the same digital economy. Table 6 summarizes the preliminary evidence corresponding to H2-H4.

Table 6. Preliminary evidence beyond platform accounts.

Question	Preliminary evidence	Interpretation
Where does e-commerce growth come from?	E-commerce value +17.08%; estimated business count +15.31%; implied nominal value per business +1.54%	Much of the aggregate increase is associated with broader participation rather than a large increase in value per business
Where does that growth occur?	Approximately 98.46% of the nominal increase occurs outside the marketplace component	Marketplace-centered evidence captures only part of recent national e-commerce growth
Are marketplace use and financial records related?	Complete financial statements: 28.63% of marketplace users versus 12.25% of non-marketplace users	Marketplace participation is associated with stronger financial recordkeeping; the evidence is observational
Do other digital records show the same growth?	BPS e-commerce +17.08%; electronic-money shopping +30.47%; mobile-banking payments and purchases +82.84%; QRIS +186.98%	Different records produce substantially different descriptions of digital growth

These comparisons are deliberately kept separate. The business-count decomposition is arithmetic rather than causal, marketplace participation is treated as an association rather than a cause of formalization, and payment values are not interpreted as alternative estimates of e-commerce sales.

The evidence nevertheless shows why measurement choice matters. Recent e-commerce growth is concentrated differently depending on whether the analyst looks at marketplace activity, national e-commerce statistics, or payment records.

5.3 Feasibility and Scope of the Preliminary Findings

The preliminary evidence is sufficient to motivate the full analysis. H1 is supported by substantial movement between transaction value and revenue within platforms. H2 and H3 can be examined using national and business-level BPS evidence, while the contrast between platform, e-commerce, and payment records provides a direct test of H4.

A separate comparison of eight platform businesses outside Indonesia provides additional evidence that the transaction-revenue boundary is not unique to the Indonesian setting. It is used as corroboration rather than added to the Indonesian sample.

Indonesia's marketplace reporting regime provides an institutional application of the same distinction. PMK 37/2025, administered by the Directorate General of Taxes (DJP), shows how seller-linked transaction records can enter reporting and withholding, but implementation begins only on 1 November 2026, so post-implementation outcomes are not yet available. Because the regime designates marketplace operators, its reach is bounded by the marketplace channel, which carried 15.8 percent of 2024 e-commerce value and 1.5 percent of 2023-2024 growth. The regulation is therefore examined as institutional architecture, not as evidence of unpaid tax or of a completed compliance effect.

At this stage, the preliminary results establish that the proposed measurement can be constructed, followed through time, connected to disclosed accounting changes, and compared with independent records of Indonesian digital activity. The remaining analysis tests how robust those conclusions are to alternative evidence and measurement choices.

6. Why the Problem Matters

The problem is not that these records differ; they are designed to measure different things. The problem arises when one is used as a substitute for another. Platform revenue measures what the platform recognizes as its own income, transaction value measures the commerce it processes, national e-commerce statistics describe a broader market, and payment systems record the movement of funds. When these measures move differently, the choice of record can change conclusions about economic scale, growth, and monetization.

Administrative use is a separate question. Platform-held transaction records may contain information about sellers and transactions that does not appear in the platform's corporate revenue,

but those records become useful to government only when identity, reporting, transmission, and matching are in place. PMK 37/2025 illustrates this institutional link without establishing a compliance or tax-revenue effect.

The thesis therefore treats measurement and administrative visibility as related but distinct problems.

7. Limitations and Robustness

The evidence supports several conclusions, but the interpretation of each result is deliberately bounded. Table 7 pairs each supported conclusion with what it does not establish.

Table 7. Main inference boundaries.

Evidence supports	Does not by itself establish
W, E, and D describe the size and movement of the transaction-revenue boundary.	Participant profit, taxable income, unpaid tax, tax evasion, or missing GDP.
Separate evidence types show how directly each platform measure represents Indonesia.	That direct, broader-scope, and reconstructed observations are interchangeable.
Disclosed revenue components can reconcile selected movements in the wedge.	A causal effect of incentives, monetization, or accounting choices.
BPS evidence describes aggregate e-commerce growth and recordkeeping associations.	Causal business entry, productivity effects, or marketplace-induced formalization.
Payment statistics and PMK provide additional measurement and institutional evidence.	Equivalent measures of sales or a completed compliance or tax-revenue effect.

Robustness analysis tests whether the main findings depend on one preferred construction. The longitudinal results are compared under narrower evidence-admission rules, alternative revenue definitions, different publication versions, and alternative assumptions for reconstructed country measures. These checks are reported as sensitivity analyses rather than statistical confidence intervals.

8. Next Steps

The main data and empirical framework are already assembled. The remaining work is to complete the robustness checks, finalize the treatment of the different evidence types, verify source consistency, and develop the preliminary results into the full thesis analysis. Table 8 sets out that schedule.

Table 8. Planned work toward the final thesis.

Period	Planned work
Proposal stage	Incorporate feedback from the proposal examination and finalize the empirical specification.
October-November 2026	Complete remaining source checks and robustness analysis; finalize the main tables and figures.
Before the final defense	Complete the full manuscript, integrate supporting evidence where it strengthens the argument, and conduct the final citation and consistency review.

The proposed study begins from a simple distinction between the value a platform processes and the revenue it recognizes. The preliminary evidence indicates that this boundary is economically large, changes through time, and produces different conclusions when compared with other records of digital activity. The final thesis will test how robust those conclusions remain under alternative evidence and measurement choices.

Appendix A - Data Sources by Evidence Type

Table A1. Source lineage by evidence type.

Evidence type	Platform / source	Transaction-value source	Revenue / comparison source
Direct Indonesia-aligned	Tokopedia e-commerce	GoTo annual report, e-commerce segment metrics	GoTo annual report, segment note
Company-reported broader scope	Blibli third-party business	Global Digital Niaga prospectus and results releases	Same issuer filings
Company-reported broader scope	Bukalapak Group	Bukalapak annual and sustainability reports	Same issuer filings
Indonesia reconstruction	Grab Indonesia	Derived from Indonesia revenue and Group monetization rate	Grab Form 20-F
Indonesia reconstruction	Shopee Indonesia	Momentum Works Southeast Asia e-commerce estimate	Derived from Sea Limited Form 20-F
National statistics	BPS-Statistics Indonesia	E-Commerce Statistics 2023 and 2024; marketplace study	Business-count and recordkeeping evidence
Payment statistics	Bank Indonesia	Payment System Statistics and QRIS reports	Growth-rate comparison only
External corroboration	Eight platform businesses	Issuer transaction measures	Issuer revenue measures
Regulatory	PMK 37/2025; DJP	Ministry of Finance / Directorate General of Taxes	Institutional architecture only

Table A1 traces each evidence type to its sources. Every figure used in this proposal traces to a named file and locator in the accompanying data package.

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